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University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

LAB REPORT

Experiment Number: 05

Experiment Title: Designing of an RC Remote Controller-Based Robot.

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Date of Submission: 13.10.2026

1. Experiment Name

Designing of an RC Remote Controller-Based Robot Using Arduino UNO

2. Objective

The main objective of this experiment is to design and implement a remote-controlled robot using an Arduino UNO, motor driver, and wireless remote controller. The robot can move forward, backward, left, and right according to commands received from the remote controller.

3. Apparatus / Hardware & Software Requirements

- List all required tools and components:

Hardware Requirements:

- Arduino UNO
- RC remote controller and IR receiver
- L293D motor driver
- Four DC geared motors
- Robot chassis and wheels
- Battery and power supply
- Jumper wires
- Breadboard (if required)

Software Requirements

- Arduino IDE
- Tinkercad Circuits (for simulation/design)
- Arduino programming language (C/C++)

4. Circuit Diagram / Schematic

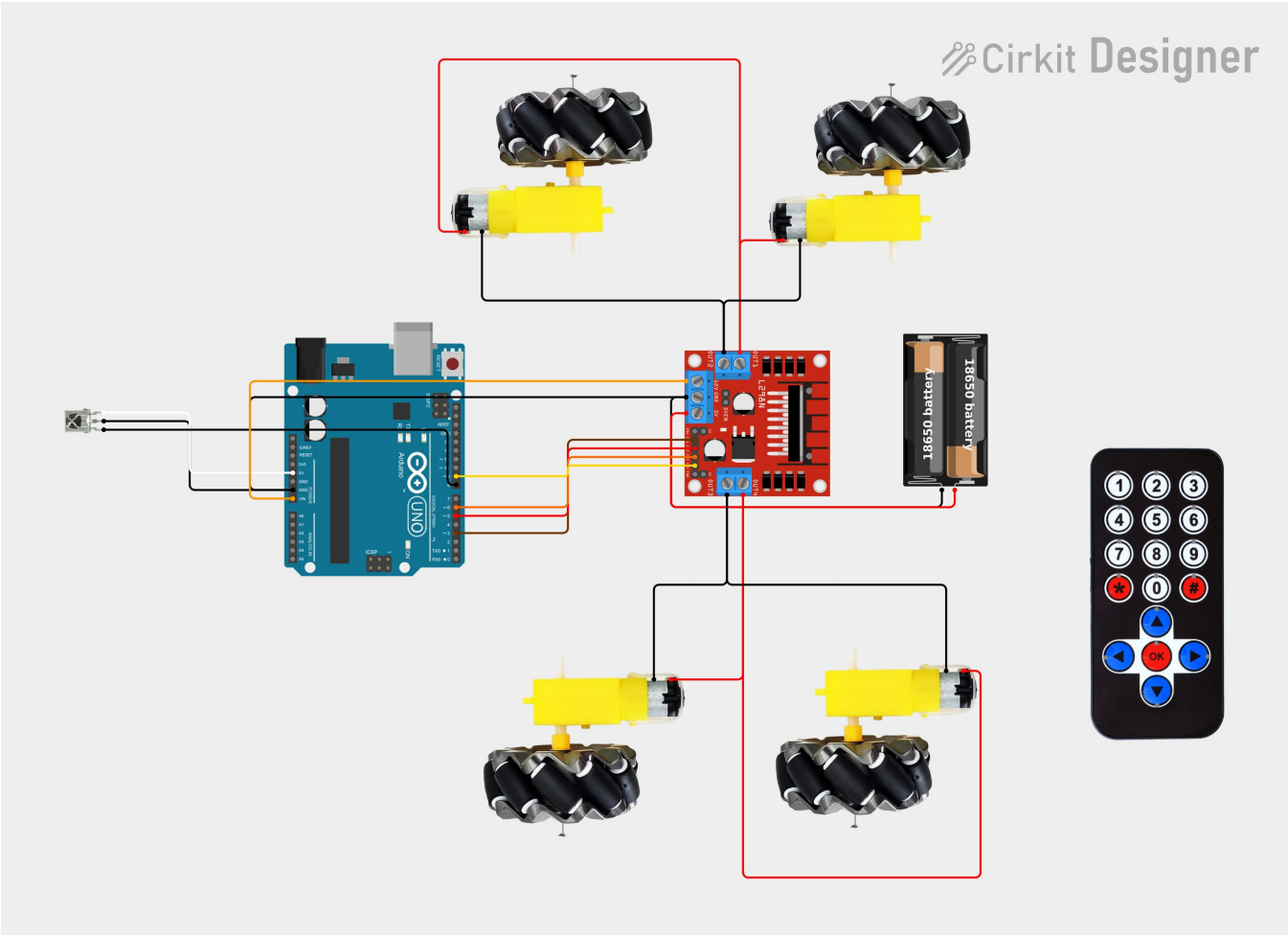


Fig: RC Remote Controller-Based Robot

5. Code / Assembly Program

```
/*
  Experiment: Remote-Controlled Robot
  Board: Arduino UNO
  Motor Driver: L293D
  Wireless Control: HC-05 Bluetooth Module

  Commands:
  F = Forward
  B = Backward
  L = Left
  R = Right
  S = Stop
*/

#define ENA 5
#define ENB 6

#define IN1 7
#define IN2 8
#define IN3 9
#define IN4 10

char command;

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);

  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  Serial.begin(9600);

  analogWrite(ENA, 180);
  analogWrite(ENB, 180);

  stopRobot();
}

void loop() {
  if (Serial.available() > 0) {
    command = Serial.read();

    switch (command) {
      case 'F':
      case 'f':
```

```
        forward();
        break;

      case 'B':
      case 'b':
        backward();
        break;

      case 'L':
      case 'l':
        left();
        break;

      case 'R':
      case 'r':
        right();
        break;

      case 'S':
      case 's':
        stopRobot();
        break;
    }
  }
}

void forward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void backward() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}

void left() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}

void right() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
```

```
}

void stopRobot() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}
```

6. Output / Observations

After uploading the program and establishing the wireless connection, the robot responds to the movement commands.

Observed Behavior:

Input Command	Robot Action
F	Move forward
B	Moves backward
L	Turn left
R	Turns right
S	Stops

Working Procedure

1. Assemble the robot chassis and attach the two DC motors and wheels.
2. Connect the motors to the L293D motor driver.
3. Connect the motor driver input and enable pins to the Arduino UNO.
4. Connect the wireless control module according to its specifications.
5. Upload the Arduino program using Arduino IDE.
6. Send movement commands from the compatible remote-control interface.
7. Observe the robot's movement in each direction.

8. Result

The remote-controlled robot was designed using an Arduino UNO and an L293D motor driver. The system demonstrates forward, backward, left, right, and stop operations in response to wireless

commands. Therefore, the objective of the experiment was achieved, provided the hardware connections and wireless control interface are configured correctly.

9. Conclusion

In this experiment, a wireless remote-controlled robot was designed and implemented. The project helped develop practical knowledge of Arduino programming, motor-driver interfacing, wireless communication, and DC motor control. It also demonstrated how wireless commands can be used to control the movement of a robotic vehicle.